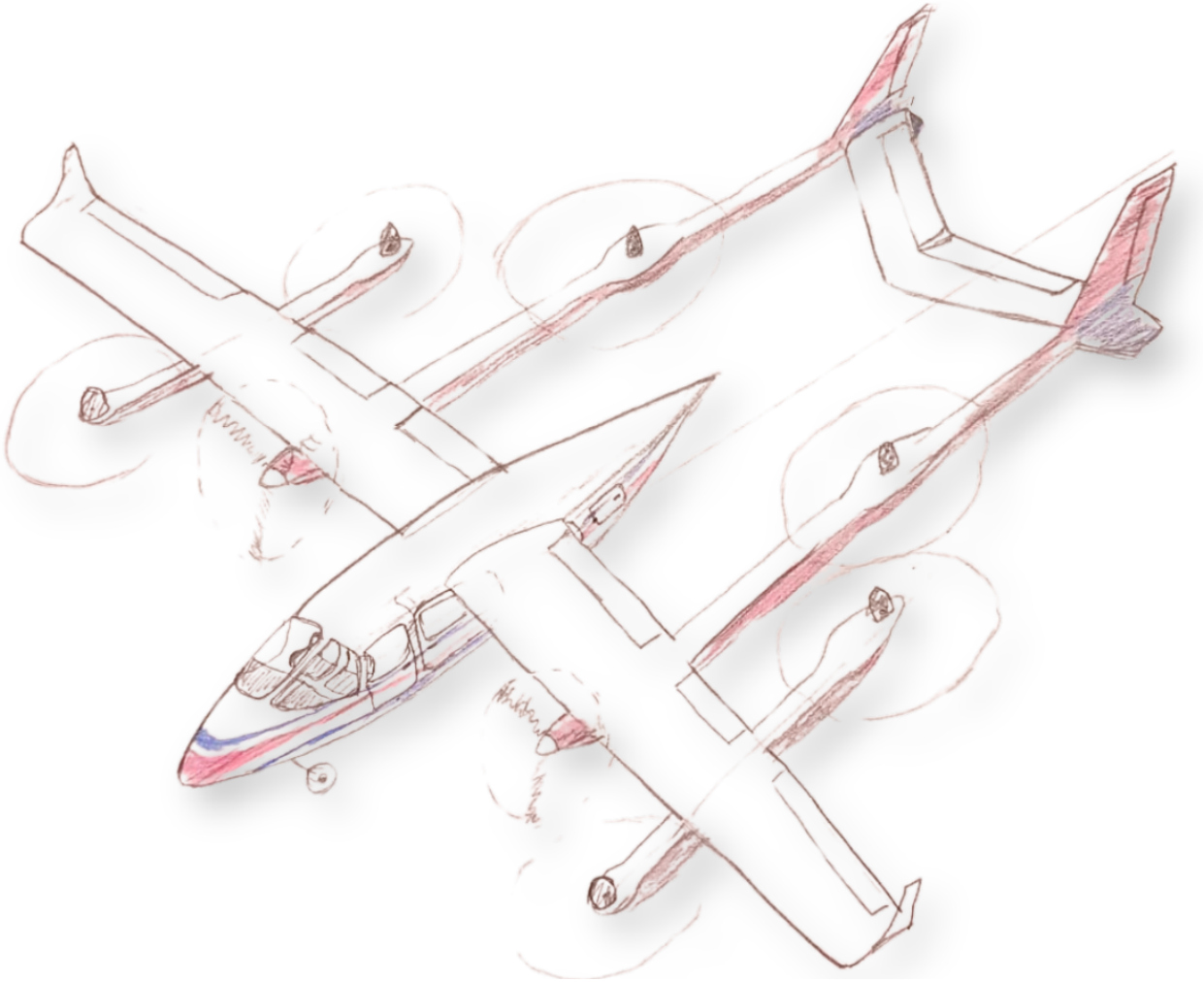
Report II

Mission Requirements and Initial Sizing



March 27, 2024

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Executive Summary

The development of the electric Vertical Takeoff and Landing (eVTOL) aircraft represents a groundbreaking approach to urban air mobility, focusing on creating an aircraft that combines the vertical lift capabilities of a helicopter with the efficient flight characteristics of an airplane. This project aims to introduce a versatile, efficient, and environmentally friendly mode of air transportation that can operate within urban areas and beyond, significantly reducing travel time for short to medium distances.

The core mission driving this project is to develop an aircraft that is not only versatile and efficient but also adaptable for various uses such as passenger transport, cargo delivery, and other specialized aerial services. Also the airplane can be modified in the future to fly in STOL and CTOL missions. The aircraft is designed with a six-seat configuration, allowing it to cater to small groups of passengers or cargo, making it ideal for a wide range of applications from commuter travel to logistics.

A crucial aspect of the aircraft's design is its modularity. This means that the aircraft can be easily adapted or modified to meet different operational needs or to incorporate new technologies as they become available. This design philosophy ensures that the aircraft remains at the forefront of aviation technology, providing sustainable and efficient solutions for urban mobility.

A significant emphasis is placed on minimizing the environmental impact of the aircraft. By using electric power, the eVTOL aims to achieve zero emissions, contributing to cleaner air and less noise pollution in urban areas. This commitment to sustainability reflects a broader effort to align with global environmental standards and to promote green aviation practices.

Several key analyses have been conducted to optimize the eVTOL design, including ground effect analysis, requirement analysis, and configuration analysis. Ground effect analysis is crucial for understanding how the aircraft performs during takeoff and landing, particularly in the complex airflows found in urban environments. Requirement analysis has

helped define the essential characteristics needed for the aircraft to meet its operational goals, such as vertical takeoff and landing capabilities, electric propulsion, and modularity. Configuration analysis led to the exploration of various design options, evaluating their feasibility, performance, and compliance with mission requirements. This rigorous analysis process has been instrumental in refining the aircraft design, ensuring that it combines optimal performance with environmental sustainability.

After thorough review and analysis, a specific design configuration was selected for further development. This configuration balances performance, efficiency, and adaptability, making it well-suited to meet the diverse needs of urban air mobility. Additionally, careful consideration has been given to patent and intellectual property issues, ensuring that the selected design respects existing patents while fostering innovation.

Risk management has been a critical component of the project, with a comprehensive analysis of potential risks and their mitigation strategies. This includes evaluating the likelihood and severity of risks related to regulations, technology development, funding, manufacturing capabilities, modularity, proof-of-concept realization, market trends, and certification. By proactively addressing these risks, the project aims to ensure the successful development and implementation of the eVTOL aircraft.

1 Mission Statement

The mission is to develop a modular aircraft that encapsulates versatility and efficiency, catering to a spectrum of aviation needs from eVTOL, STOL to CTOL capabilities. The aircraft's design will feature a consistent four-seat airframe, ensuring a broad operational range of 200 km across various configurations. This modularity aims to provide seamless adaptability for diverse applications including commuter travel, cargo transportation, and aerial services.

In adherence to the principles of green aviation, the design will be anchored in the application of emerging environmentally friendly technologies. The commitment to sustainability will be reflected in the reduction of the aircraft's carbon footprint, promoting energy efficiency and lower emissions in compliance with global environmental standards.

The vision for this aircraft is to set a benchmark in the aviation industry for flexible design while upholding stringent environmental directives. It will exemplify the integration of innovative modular engineering with the values of eco-consciousness, paving the way for a new generation of aircraft that align with the imperatives of future-forward and sustainable aviation.

2 Operational Concept

2.1 Basic

For this category, the operations are the ones not included in the enhanced version, so everything that is different from Urban Air Mobility (UAM) or the Air Taxi concept. This means but is not limited to General Aviation, Special Operation, manned cargo and outside congested areas. It could also be classified by the number of passengers, for basic: 0 to 9 passengers. For this category, in the event of the loss of an engine, it would be necessary to complete a controlled emergency landing (CEL). For this type of operation, the trajectory shown in Figure 2.2 is proposed, so in case of failure the pilot could perform a landing in a

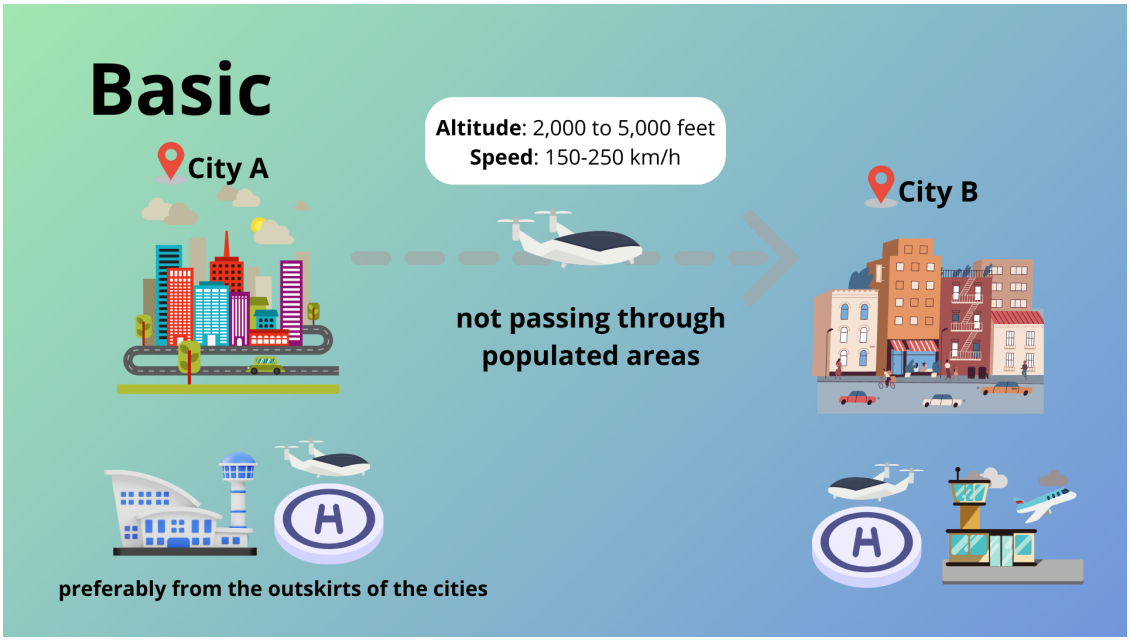


Figure 2.1: Concept for a connection between cities not passing through congested areas

short runway as a STOL.

2.2 Enhanced

As mentioned before, the enhanced category include all the UAM passing through congested areas and air taxi service. It is classified for flights from 10 to 19 passengers. In case of a failure, it must perform a continuous safe flight and landing (CSFL), meaning it should perform the same original trajectory having multiple redundant systems or at least find another vertiport that where it could make the safe vertical landing. The same profile from the basic operation is proposed, but with the consideration that the vertiports should be at higher buildings, so there are no urban obstacles in the climbing stage.

3 Ground Effect Analysis

- **Suck-Down Effect:** A critical phenomenon for VTOL vehicles, caused by the acceleration of the air mass around the vehicle due to viscosity, creating a downward

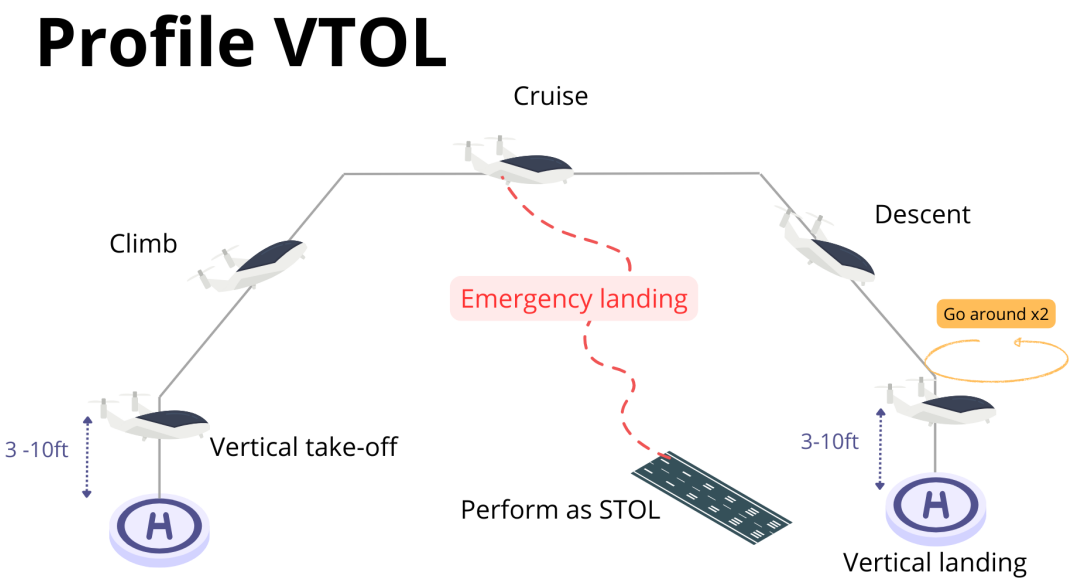


Figure 2.2: Flight profile for basic operation

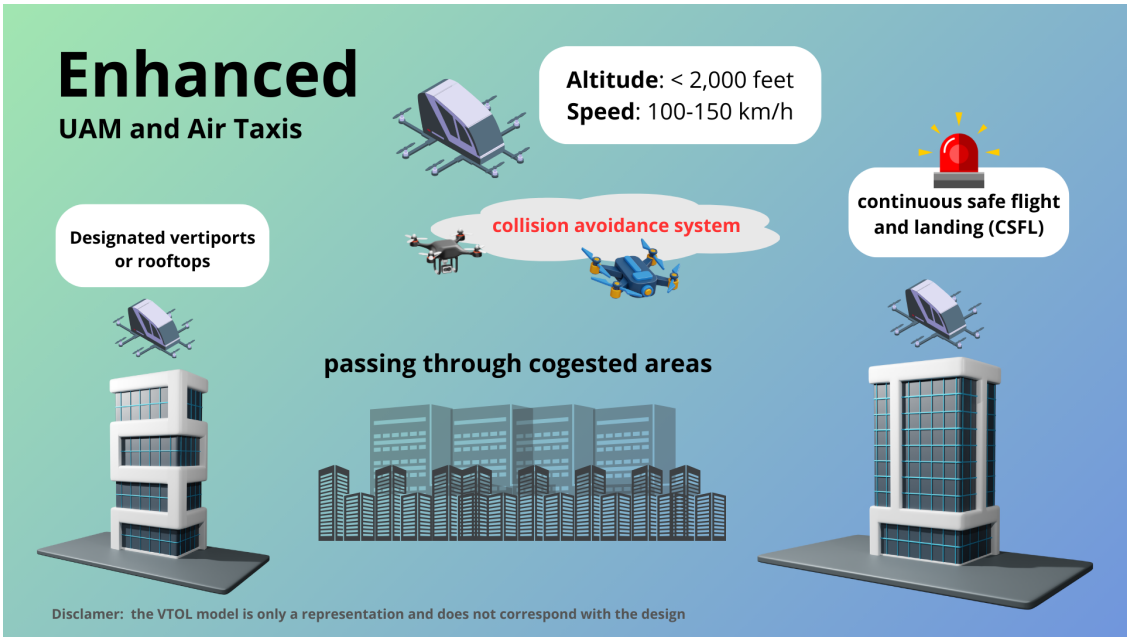


Figure 2.3: Concept for urban operation

flowfield. This effect is intensified near the ground due to the improved mixing of the deflecting jet exhaust and adjacent air, which can represent a "vertical drag" of 2-6% of T/W in free air and up to 30% of T/W in ground effect.

- **Ground Effect:** An aerodynamic phenomenon that occurs when a vehicle is close to the ground, resulting in increased lift efficiency due to the restriction of wingtip vortices. This significantly affects VTOL vehicle operations, especially during takeoff and landing, by reducing induced drag and increasing thrust efficiency. This effect is shown in figure 3.1.

When propellers or rotors are used to provide necessary lift in VTOL vehicles, their placement is crucial because any part of the structure affected by the rotor's wake produces a download, reducing net lift. The estimation given for VTOL vehicles using propellers indicates that the ground effect can increase the "vertical drag" up to 30% of T/W, highlighting the importance of considering this effect in the design and operation of VTOL vehicles [1].

If rotors are used to provide the necessary lift, their placement has to be carefully considered. Any portion of the airframe surface impinged by the rotor's wake produces a download, which reduces the net lift. The US Army Material Command's Helicopter Design Handbook [2] gives the following equation to approximate this effect:

$$\text{Download}_{\text{Hover}} = \frac{1}{2} * \frac{T}{S_{\text{Disk}}} * S_{\text{exposed to wake}}$$

4 Requirement Analysis

The following requirements are assessed for the proposed configurations:

- VTOL for enhanced operations: This determines that the aircraft will operate in urban environments. Low noise, small aircraft footprint, and high safety standards.
- Electric propulsion system.

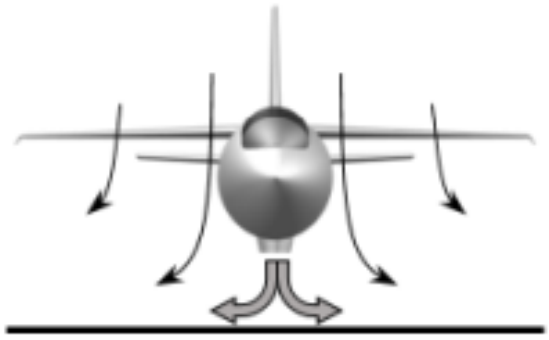


Figure 3.1: Ground Effect

- Modularity design for further development into STOL and CTOL aircraft
- Spacious cabin with easy access and evacuation for 6 people.
- 100km in range.
- Automatic production consideration.

5 Configuration Analysis

9 Configurations were initially considered and 5 were pre-selected by the team to comply with the requirements. In the following section, each one of the configurations is shown with their respective parameters. Also, some weight, power, and energy estimations are made to predict the general sizing of the aircraft. This was done with our developed software using the equations and methods presented in the works of [3], [4], [5], [6] and [7].

5.1 Design 3

Double boom configuration (patent available) with 4 VTOL motors and 2 cruise motors. All fixed. H tail and big fuselage and door for payload. The main design parameters are shown

in Table 5.1, the sketch can be seen in figure 5.1 and the main weights, power and energy requirements are summarized in figure 5.2.

Table 5.1: Design 3 parameters

Patent state	Available	Total motors	6
Wingspan $[m]$	15	Motors used for VTOL	4
Mean chord $[m]$	1.3	Total power $[kW]$	924
Wing area $[m^2]$	20	Power per motor $[kW]$	231
Fuselage length $[m]$	6	Total power $[HP]$	1216
Seats	6	Power per motor $[HP]$	304

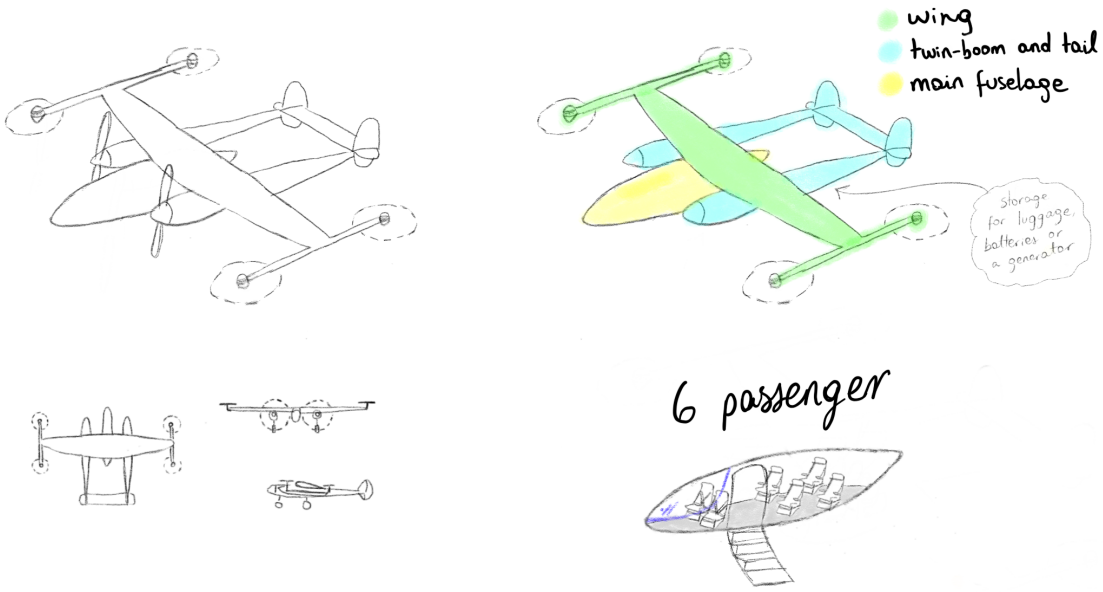


Figure 5.1: Design 3 sketch

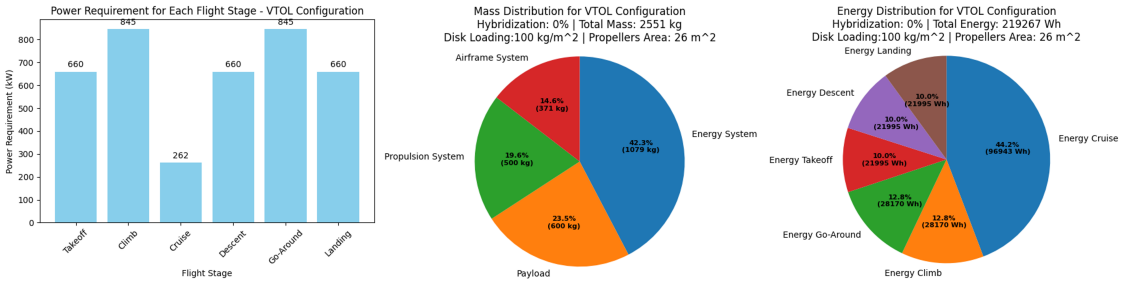


Figure 5.2: Design 3 analysis

5.2 Design 5

Inspired by the design shown in figure 5.3, the following configuration is considered: Double boom configuration with 6 VTOL motors and 2 cruise motors. All fixed. H tail and big fuselage and door for payload. The main design parameters are shown in Table 5.2, the sketch can be seen in figure 5.4 and the main weights, power and energy requirements are summarized in figure 5.5.

Table 5.2: Design 5 parameters

Patent state	Infringement US9120560B1	Total motors	8
Wingspan $[m]$	15	Motors used for VTOL	6
Mean chord $[m]$	1.3	Total power $[kW]$	956
Wing area $[m^2]$	20	Power per motor $[kW]$	159
Fuselage length $[m]$	8	Total power $[HP]$	1258
Seats	6	Power per motor $[HP]$	210

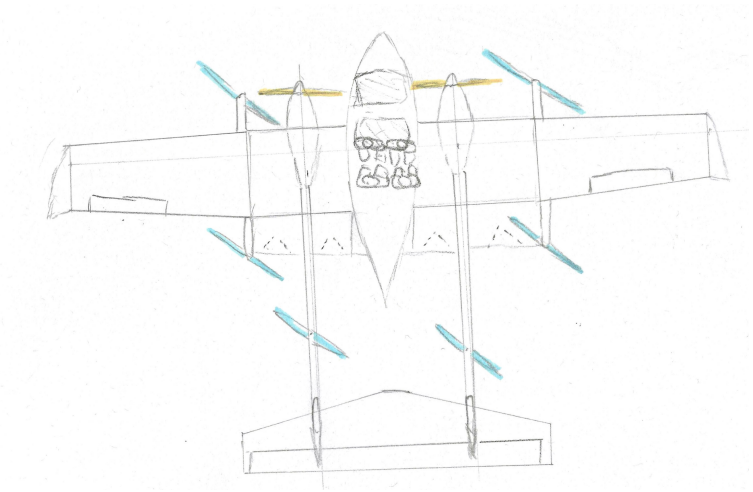


Figure 5.3: Concept for Design 5

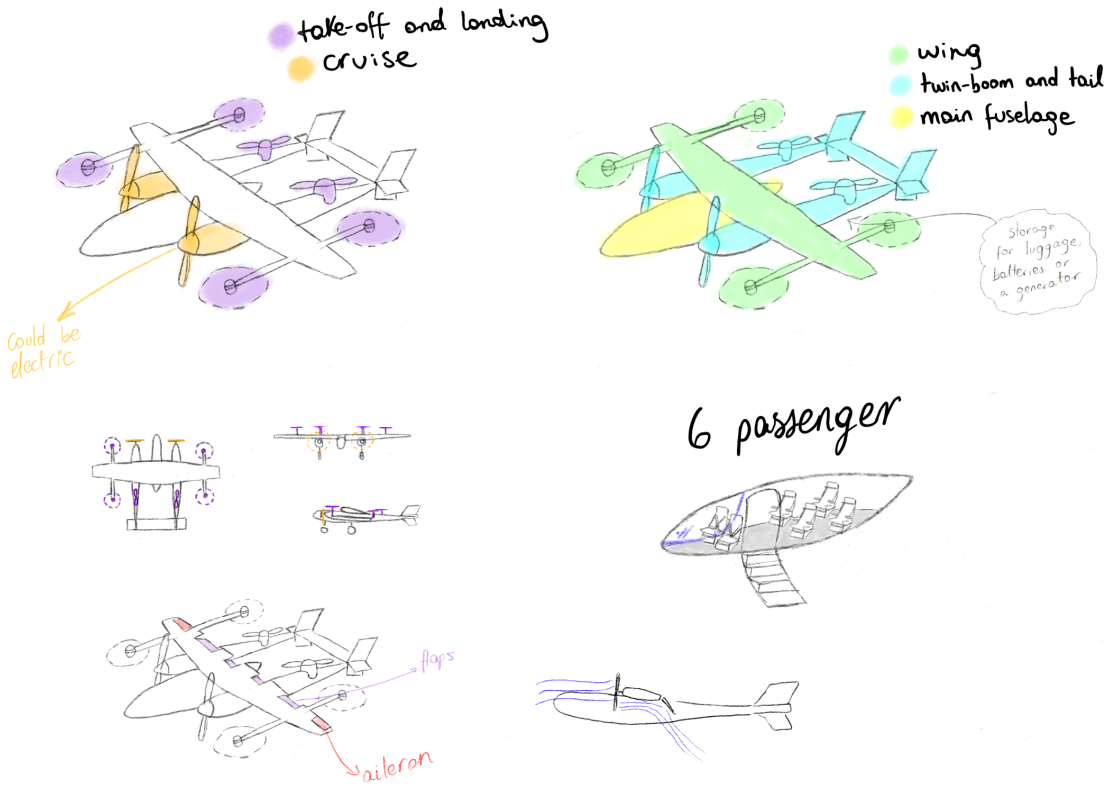


Figure 5.4: Design 5 sketch

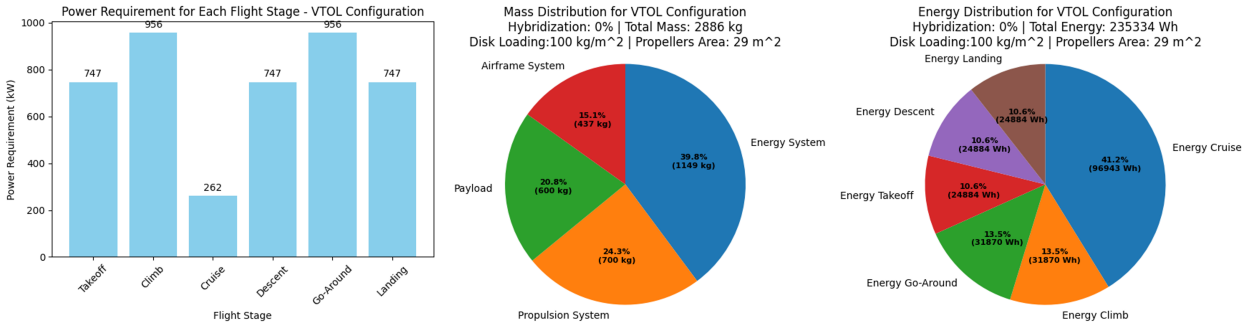


Figure 5.5: Design 5 analysis

5.3 Design 6

A high wing design with 2 tilt rotors (external ones) and 6 fixed motors (only for vertical stages). T tail for control authority. Big fuselage area and door. Modularity in tail design. The main design parameters are shown in Table 5.3, the sketch can be seen in figure 5.6 and the main weights, power and energy requirements are summarized in figure 5.7

Table 5.3: Design 6 parameters

Patent state	Available - US20200391859A	Total motors	8
Wingspan $[m]$	15	Motors used for VTOL	8
Mean chord $[m]$	1	Total power $[kW]$	944
Wing area $[m^2]$	15	Power per motor $[kW]$	118
Fuselage length $[m]$	10	Total power $[HP]$	1242
Seats	6	Power per motor $[HP]$	155

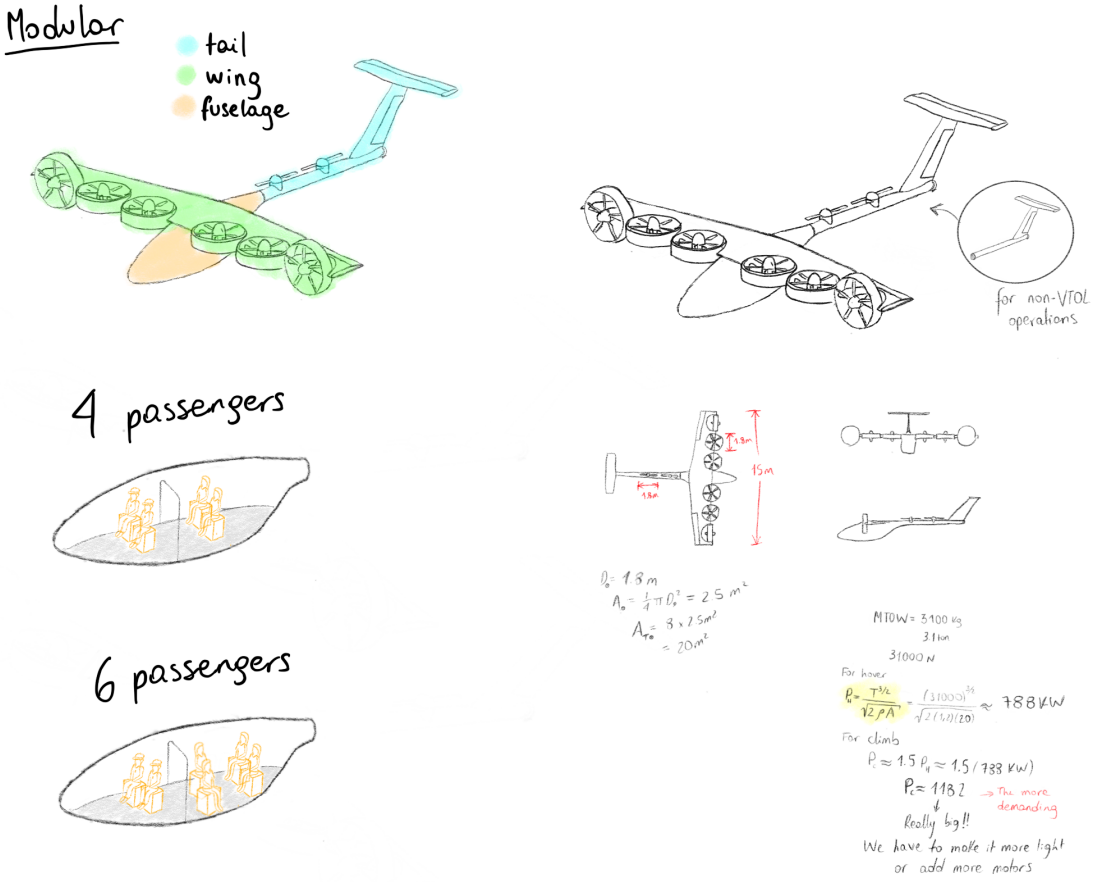


Figure 5.6: Design 6 sketch

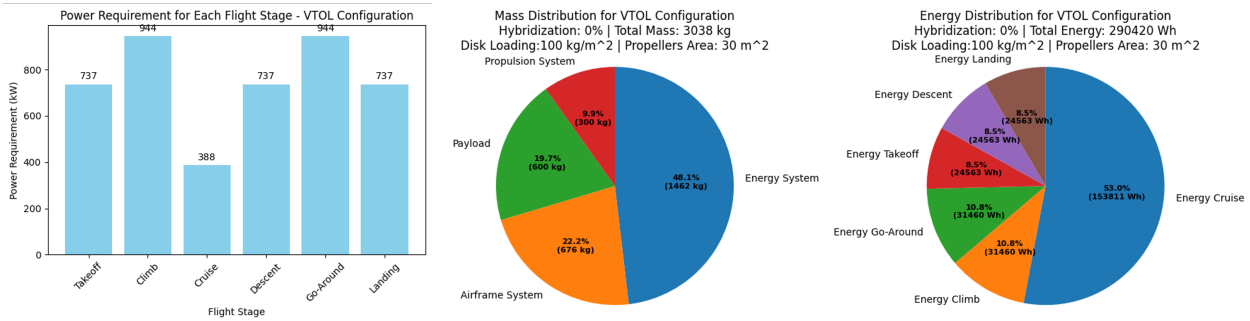


Figure 5.7: Design 6 analysis

5.4 Design 7

A tandem wing design that allows for a wider center of gravity range, which allows passengers to use a wide door for easy boarding and deboarding. With a 4-tilt rotor system. This design has the goal of providing as much lift as possible with the available fixed wing surfaces and distributing the weight over the triple fuselage. The main design parameters are shown in Table 5.4, the sketch can be seen in figure 5.8 and 5.9. The main weights, power and energy requirements are summarized in figure 5.10.

Table 5.4: Design 7 parameters

Patent state	Available - US20200140079A1	Total motors	4
Wingspan $[m]$	12	Motors used for VTOL	4
Mean chord $[m]$	1.3	Total power $[kW]$	1062
Wing area $[m^2]$	31.2	Power per motor $[kW]$	266
Fuselage length $[m]$	9	Total power $[HP]$	1397
Seats	6	Power per motor $[HP]$	349

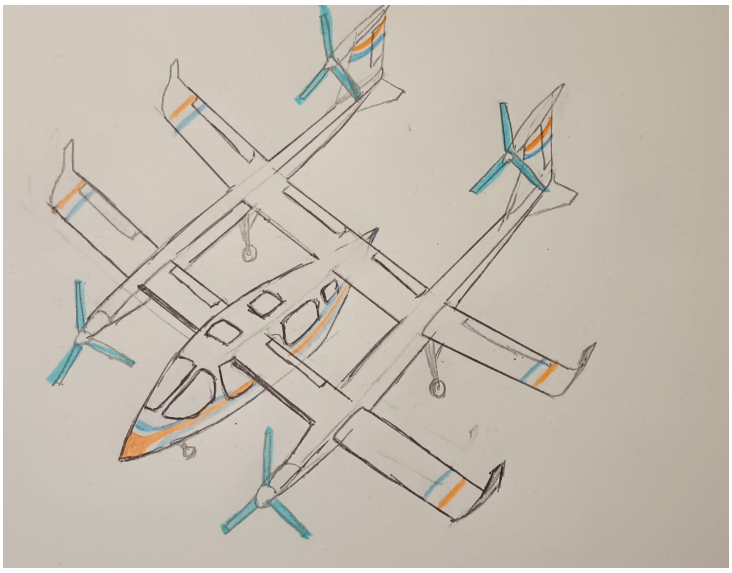


Figure 5.8: Design 7 sketch

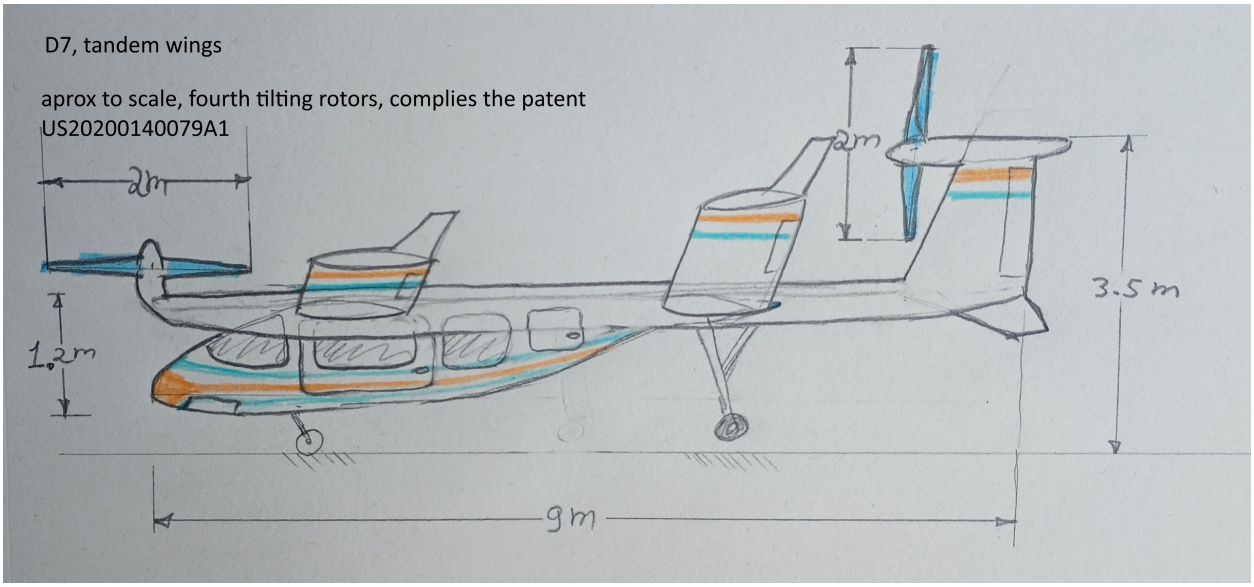


Figure 5.9: Design 7 general dimensions

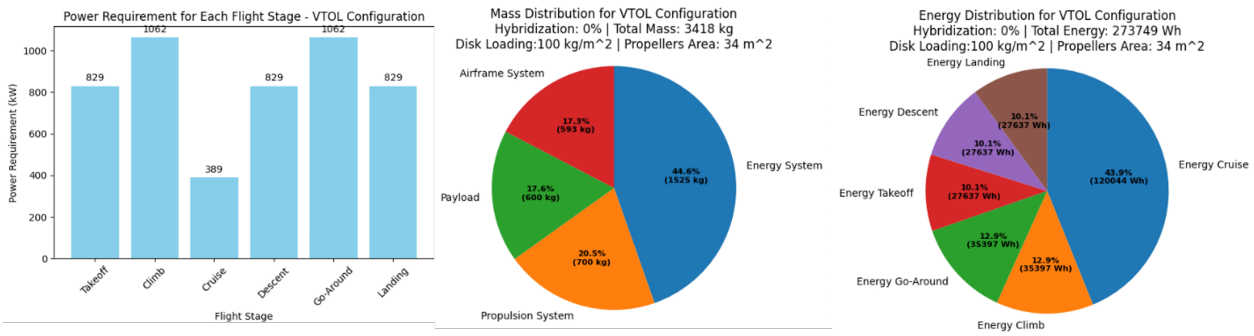


Figure 5.10: Design 7 analysis

5.5 Design 8

High back wing (4 tilted rotors and 1 fixed pusher) with canard configuration (distributed ducted propulsion). Big fuselage area for six passengers. The main design parameters are shown in Table 5.5, the sketch can be seen in figure 5.11 and the main weights, power and energy requirements are summarized in figure 5.12

Table 5.5: Design 8 parameters

Patent state	Available	Total motors	13
Wingspan $[m]$	10	Motors used for VTOL	12
Mean chord $[m]$	1.36	Total power $[kW]$	911
Wing area $[m^2]$	13.58	Power per motor $[kW]$	76
Fuselage length $[m]$	6.9	Total power $[HP]$	1199
Seats	6	Power per motor $[HP]$	100

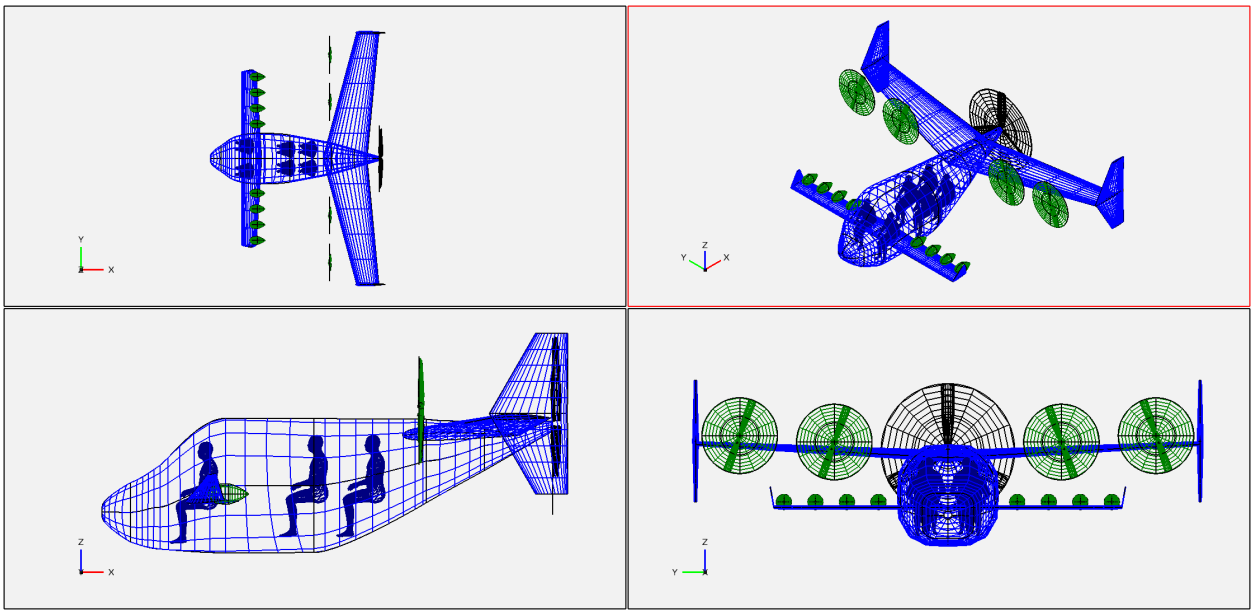


Figure 5.11: Design 8 sketch

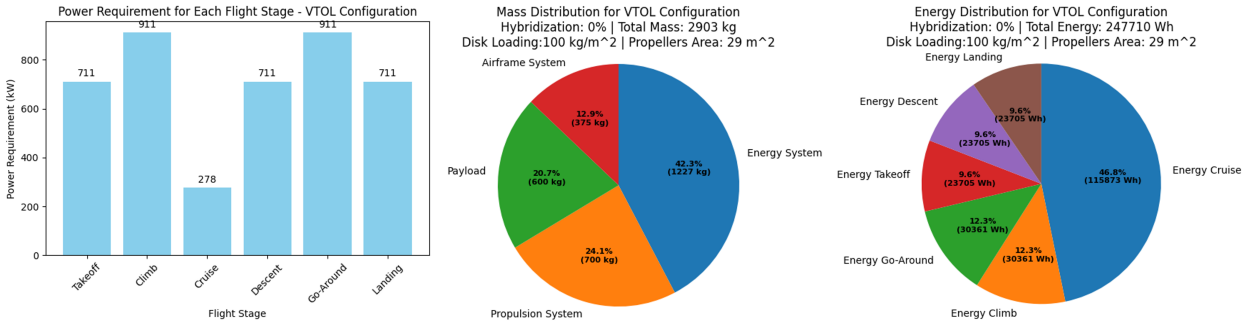


Figure 5.12: Design 8 analysis

6 Configuration Selection

After the committee review with Mr Radim Olbrecht, Leos Vybiral and Felipe Correa, of the proposals on March 26, 2024, the selected design to continue the design project is D5. The selected configuration is shown in figures 5.3 and 5.4. Furthermore, the following figures 6.1 and 6.2 show the eVTOL and eSTOL configurations. These proposed designs are for display only, showing the modularity of the design and its ability to have different missions.

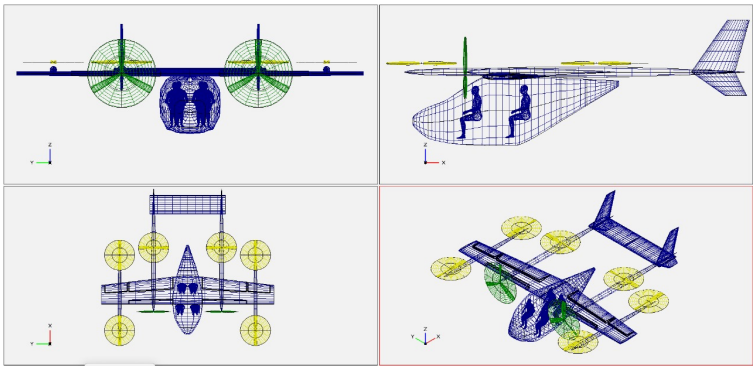


Figure 6.1: Proposed eVTOL 4 seater

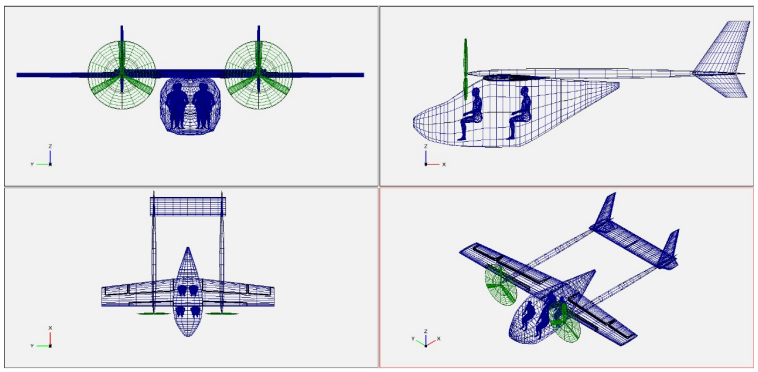


Figure 6.2: Proposed eSTOL 4 seater

7 Patent Notes

During the conceptual design phase, the project was developed with patent and intellectual property restrictions in mind.

To get a first iteration of this restriction, a search on google patents and USPTO (United States Patent and Trademark Office) with keywords such as: VTOL, aircraft.

The search returns 44,557 results, including worldwide coverage.

Then a deeper search using keywords such as VTOL aircraft vertical takeoff and landing will narrow the result to 32,534.

Including the word tiltrotor reduces the results to 4,756.

Some results for abandoned patents include the following, which were suggested as possible design guidelines:us20200140079a1 from Bell Helicopters Textron.

In tables 7.1 and 7.2 the preliminary analysis of the potential patent infringement was summarized related to the proposed designs.

With the design selected, the patents shown in figure 7.1 can have some similarity with the configuration.

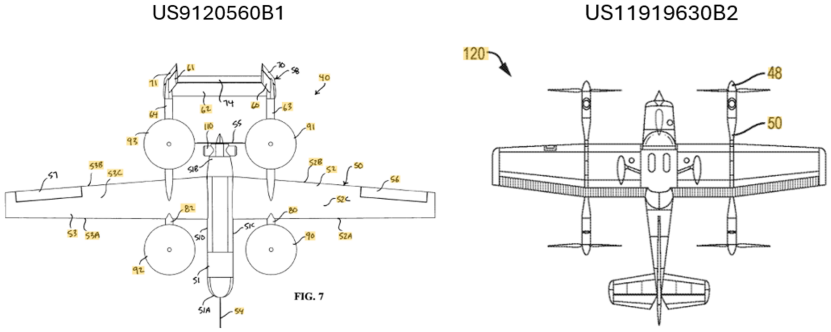


Figure 7.1: Potential patent infringements

Table 7.1: Designs preliminary patent evaluation part 1

	Drawing 1	
Item	4 seats	6 seats
Comply with FAA mosaic	YES	NO
Multiporpuse Fuselage	YES	
Grow to 6-9 places	YES	
Patent and available design PI	YES	
Notes	We suggest more searching of patents status, looks available	
	Drawing 2	
Item	4 seats	6 seats
Comply with FAA mosaic	YES	NO
Multiporpuse Fuselage	YES	
Grow to 6-9 places	YES	
Patent and available design PI	NO	
Notes	Review Patent US11919630B2	
	Drawing 3	
Item	4 seats	6 seats
Comply with FAA mosaic	YES	NO
Multiporpuse Fuselage	YES	
Grow to 6-9 places	YES	
Patent and available design PI	YES	
Notes	We suggest more searching of patents status, looks available	
	Drawing 4	
Item	4	6 seats
Comply with FAA mosaic	YES	NO
Multiporpuse Fuselage	YES	
Grow to 6-9 places	YES	
Patent and available design PI	NO	
Notes	Review Patent US20190061968A1	

Table 7.2: Designs preliminary patent evaluation part 2

	Drawing 5	
Item	4 seats	6 seats
Comply with FAA mosaic	YES	NO
Multiporpuse Fuselage	YES	
Grow to 6-9 places	YES	
Patent and available design PI	NO	
Notes	Infringement US9120560B1	
	Drawing 6	
Item	4seats	6 seats
Comply with FAA mosaic	YES	NO
Multiporpuse Fuselage	YES	
Grow to 6-9 places	YES	
Patent and available design PI	**YES**	
Notes	Abandoned Patent US20200391859A	
	Drawing 7	
Item	4seats	6 seats
Comply with FAA mosaic	YES	NO
Multiporpuse Fuselage	YES	
Grow to 6-9 places	YES	
Patent and available design PI	**YES**	
Notes	Abandoned Patent US20200140079A1	

8 All-electric and Hybrid Analysis

All-electric eVTOL aircraft present a mix of advantages and challenges rooted in electric propulsion and battery technology. Their electric motors offer high efficiency, reduced maintenance, and quieter operations, aligning with environmental goals by eliminating emissions. However, the main drawback is the limited energy density of batteries, which restricts range and payload, making these aircraft more suitable for short-range missions. The weight of batteries also demands more power for lift-off, further limiting operational capacity. While rapid charging and safety measures for battery management are evolving, overcoming these limitations is crucial for broadening the operational scope of all-electric eVTOLs [8], [9], [10], [11].

Table 8.1: Main aircraft all-electric configuration requirements

Parameter	Units	eVTOL	eSTOL
MTOW	kg	2791	2100
Stall speed	m/s (knots)	31 (60)	31 (60)
Cruise speed	m/s (knots)	63.6 (123.6)	75 (145)
Endurance	h	0.45	0.75
Range	km	100	202
Payload	kg	600	600
Battery capacity	kWh	223	130
Battery mass	kg	1262	660
Electric motors power	kW	957	390
Motor mass	kg	320	130

Hybrid eVTOL aircraft blend electric propulsion with Internal Combustion Engines (ICE), extending range and payload beyond all-electric limitations while maintaining precise, efficient maneuvers. This configuration, however, complicates design and operation with dual power systems, leading to potential increases in maintenance and costs. While they

reduce emissions compared to ICE-only models, they do not eliminate them, diluting some environmental benefits. Despite these trade-offs, hybrid eVTOLs serve as a transitional step, offering improved performance with a nod toward sustainability [12], [13].

In table 8.2 we can see the benefit of using an increasing hybridization level (by the amount of power generated from traditional fuel). This estimation results in almost a thousand-kilogram reduction of the total weight of the aircraft capable of the same mission. The reduction of battery weight results in an airframe weight which results in even a bigger MTOW reduction.

Table 8.2: Aircraft hybrid configuration evaluation

		Hybridization level				
Parameter	Units	0%	25%	50%	75%	100%
MTOW	kg	2791	2494	2228	1987	1769
Battery capacity	kWh	223	208.7	195.5	183.5	0
Battery mass	kg	1262	884	552	259	0
Fuel mass	kg	0	43	81	115	144
Electric motors power	kW	957	856	764	682	607

9 Fuselage and Ergonomics

With the first sizing iteration, special attention was paid to the correct placement of two men side by side, as shown in figures 9.1 and 9.2.

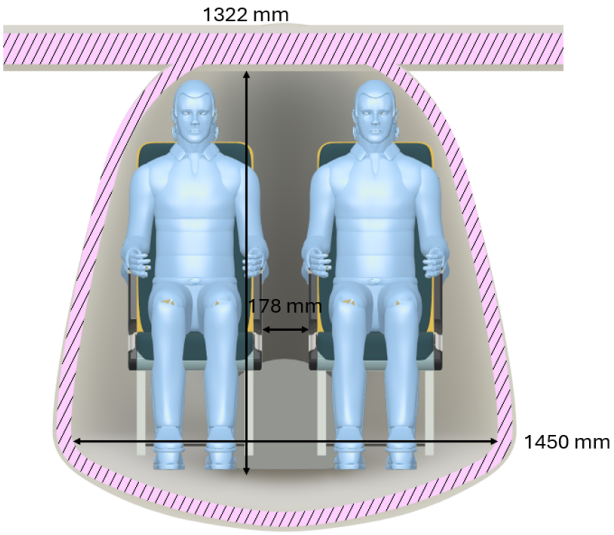


Figure 9.1: Cabin space for passengers

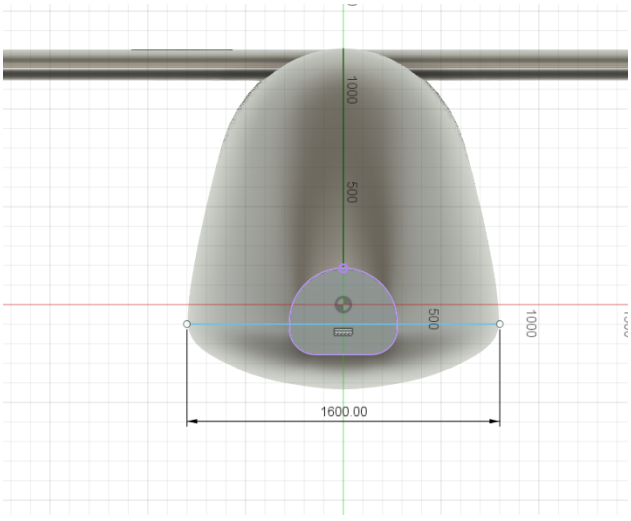


Figure 9.2: External dimension of the fuselage

10 Initial Sizing Validation

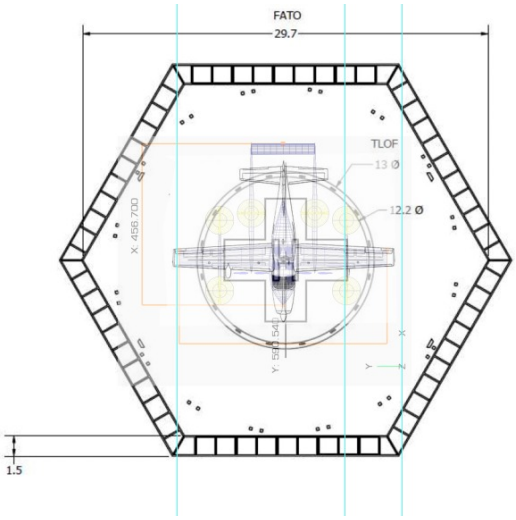


Figure 10.1: Size comparison with heliport

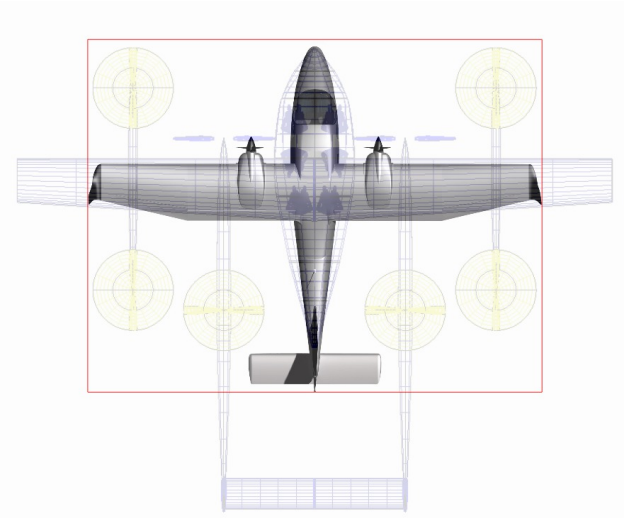


Figure 10.2: Size comparison with Tecnam p2006t

11 Risk Analysis

For the entirety of the project, the main risks were evaluated and some mitigation techniques are proposed. The likelihood and severity of the risks are evaluated following the guide of table 11.1. Then the risk is quantified following table 11.2 and then, the consequences are as shown in table 11.3. The result of this risk analysis is finally shown and summarized in table 11.4.

Table 11.1: Risks Likelihood and Severity

Level		**Likelihood**	**Severity**
5	Maximum	100%	Unacceptable
4	High	10%	Major reduction in mission objectives
3	Medium	1%	Moderate reduction
2	Low	0.1%	Moderate reduction
1	Minimum	less than 0.1%	Minimal or no impact

Table 11.2: Risk Quantification Strategy

Severity	**5**	**5**	**10**	15	20	25
	4	4	8	12	16	20
	3	3	6	9	12	15
	2	2	4	6	8	10
	1	1	2	3	4	5
		1	**2**	**3**	**4**	**5**
		Likelihood				

Table 11.3: Risks Consequences

Risk Quantification	**Consequence**
>20	Maximum disruption of project plan. Implement new process or plan
15-20	High disruption of project plan. Implement new project or plan
10-14	Some disruption of project plan. Consider alternative process
5-9	Little disruption of project plan. Management actions
<5	Little to no disruption. Mitigate with management

Table 11.4: Risks Identification

Code	Risk Name	S	L	SxL	Description	Mitigation Strategy
PL-1	Regulations change	5	2	10	Regulations change drastically in the next 5 years	Close regulation monitoring during design.
PL-2	Battery technology level not achieved	4	3	12	The energy density level of available batteries does not surpass the 400 Wh/kg	Usage of alternative energetic systems
PL-3	In-house technology development not achieved	3	3	9	The required propietary technologies are not developed successfully	Usage of comercial alternatives compromising market competitiveness
PL-4	Funding not achieved	4	3	12	Capital investors or funding opportunities are not interested in the project	Major project focus change to attract the capital necessary
PL-5	Manufacturing capabilities not as efficient as estimated	2	2	4	The engineering and production costs grow because of manufacturing limitations	Close production engineering monitoring from early stages of the design
PL-6	Aircraft modularity compromises the eVTOL mission	5	3	15	The modularity makes the project too complex and the resulting configurations are not as efficient as required	Top mission priorities definition and system-level analysis from the beginning of the project
PL-7	Proof-of-concept not achievable	4	2	8	The concept results unachievable by the deadline for the PoC	Redefinition of the PoC and resource intesification towards this objective
PL-8	Market shift in mobility trends	3	1	3	The air mobility is no longer a need for the market	Shifting focus to other market needs (cargo, military, etc)
PL-9	Certification not achievable	5	3	15	The regulators impose higher levels for certification that make the aircraft non-viable	Major aircraft design changes need to be made

12 Conclusion

The design is intended to provide 6 seats, in a configuration of 4 passengers plus a pilot and co-pilot, to comply with enhanced operations or over urban and dense population areas. It must therefore have the ability to reach a vertiport in case of emergency. In addition, the aircraft must have the capability to operate as an eStol aircraft and Ctol, which means that traditional airports are also available to operate and provide services. The aircraft in a dirt or with hiperlift devices will have in ISA conditions a stall speed projected over 60 kn approximately if only two engines are operational. In eStol conditions the stall speed will be lower, this value will be available after the next iteration of the design. The cabin interior volume and areas are considered under the regulations, so the certification is achievable with 6 adults. The cabin width is above the value of an Embraer Phenom 300 jet, so as a reference the comfort is comparable to a luxury small jet.

The iteration provided a payload comparable to a Cessna 337 with similar arrangement with twin booms, but with the center fuselage commonality of parts to allow the design to grow for future versions or toe shrink to the smallest models. The sizing taking into consideration the gross weight and power, is similar to a Cessna 208 Caravan, also the performance in stall and overall dimensions, this is a correlation that match the equations as a validation of the engineering calculations, this will be more precise during further analysis. This cabin sizing also provide capacity as CASEVAC, MEDEVAC, SAR CSAR for medical, search, rescue and related emergency services, this is a conceptual approach and will be analyzed in future engineering stages. The comparison to a heliport sized to a UH-60 Blackhawk, Augusta Westland AW139 and ASW532, so from an operational point of view. The design can operate in a similar regulation to Class 1 (a classification to operate over urban areas and with the performance to land safely even in case of failure of an engine) this will be more deeply analyzed in future engineering stages.

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